SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SIMATS ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
CSA17 - ARTIFICIAL INTELLIGENCE

Experiment No. 4: Python Program for the Crypt-Arithmetic Problem

Aim

To write a Python program to solve a Crypt-Arithmetic problem (SEND + MORE = MONEY), which involves assigning a unique digit (0-9) to each distinct letter such that the resulting arithmetic equation holds true, using a constraint-based brute-force search.

Algorithm

1. Start.

2. Identify all the distinct letters involved in the crypt-arithmetic puzzle, e.g., S, E, N, D, M, O, R, Y for SEND + MORE = MONEY.

3. Generate all possible permutations of the digits 0-9 taken as many at a time as the number of distinct letters, using the itertools.permutations function.

4. For each permutation, assign each digit to the corresponding letter to form a mapping (dictionary).

5. Apply the constraint that the leading letters of each word (S and M in this case) cannot be assigned the digit 0, since a number cannot have a leading zero.

6. Using the current mapping, compute the numeric value of SEND, MORE, and MONEY by substituting each letter with its mapped digit.

7. Check whether the equation SEND + MORE == MONEY holds true for the current mapping.

8. If the equation is satisfied, store this mapping as the valid solution and stop the search.

9. If the equation is not satisfied, discard the current permutation and continue with the next one.

10. If no permutation satisfies the equation after checking all possibilities, report that no solution exists.

11. Print the digit assigned to each letter and display the full addition with substituted digit values.

12. Stop.

Pseudocode

```
BEGIN
  letters = ['S','E','N','D','M','O','R','Y']
  FOR each permutation P of digits (0-9) taken 8 at a time DO
      mapping = ASSIGN each letter in letters a digit from P

      IF mapping['S'] == 0 OR mapping['M'] == 0 THEN
          CONTINUE   // no leading zero
      END IF

      SEND  = 1000*S + 100*E + 10*N + D
      MORE  = 1000*M + 100*O + 10*R + E
      MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

      IF SEND + MORE == MONEY THEN
          PRINT "Solution found"
```

```
            PRINT mapping, SEND, MORE, MONEY
            BREAK
        END IF
    END FOR

    IF no mapping satisfied the equation THEN
        PRINT "No solution exists"
    END IF
END
```

Result

The Python program for the Crypt-Arithmetic problem was executed successfully. The program systematically tried digit permutations for the letters in SEND, MORE, and MONEY, applied the no-leading-zero constraint, and verified the equation SEND + MORE = MONEY for each mapping. The unique digit assigned to every letter was found, and the equation was verified numerically (9567 + 1085 = 10652). Thus, the Crypt-Arithmetic problem was solved successfully using Python.

Output Screenshot:

```
Terminal — cryptarithmetic.py

$ python3 cryptarithmetic.py
Solving:   S E N D
         + M O R E
         ---------
         M O N E Y

Solution found!

S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2

  9567
+ 1085
------
 10652

Crypt-Arithmetic problem solved successfully!
```